

CS449 / CS549 Midterm 1

I. QUESTION

Given the following sentences:

```
<s> ali topu at . </s>
<s> ali ayşeye çiçek al . </s>
<s> ali topu sev . </s>
<s> ayşe topu at . </s>
<s> ali ayşeye topu at . </s>
```

- Calculate the following unigram probabilities: $P(ali)$, $P(< s >)$, $P(ayşeye)$, $P(at)$. Assume that add- k smoothing is applied with $k = 1$.
- Calculate the following bigram probabilities: $P(ali | < s >)$, $P(topu | ali)$, $P(< /s > | .)$, $P(ayşeye | ali)$. Assume that add- k smoothing is applied with $k = 2$.

II. QUESTION

Let us say we have 4 states:

S_1 : A, S_2 : C, S_3 : G, S_4 : T

with initial probabilities: $\Pi = [0.3, 0.4, 0.1, 0.2]$

and the transition matrix is:

$$\mathbf{A} = \begin{bmatrix} 0.2 & 0.2 & 0.3 & 0.3 \\ 0.1 & 0.3 & 0.2 & 0.4 \\ 0.4 & 0.4 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.7 \end{bmatrix}$$

- Give a sequence S consisting of 7 symbols, where each state occurs at least once. Calculate the probability of the sequence S .
- Give 10 different sequences $S_1, S_2, \dots, S_{10}$ of size 6, where each state occurs in each sequence at least once. Given those ten sequences estimate the initial probabilities for Π and $\mathbf{A}$.

III. QUESTION

agree.01

Arg0: Agreeer

Arg1: Proposition

Arg2: Other entity agreeing

fall.01

Arg1: Logical subject, patient, thing falling

Arg2: Extent, amount fallen

Arg3: start point

Arg4: end point, end state of arg1

increase.01

Arg0: causer of increase

Arg1: thing increasing

Arg2: amount increased by, EXT, or MNR

Arg3: start point

Arg4: end point

Annotate the following sentences with the arguments above:

- The group agreed it wouldn't make an offer
- Sales fell to \$25 million from \$27 million
- John agrees with Mary on everything.
- The price of bananas was increased again by Big Fruit Co.
- The average junk bond fell by 4.2%

IV. QUESTION

Given the following documents:

Doc1: ali topu at

Doc2: ali ayşeye çiçek al

Doc3: ali topu sev

Doc4: ayşe topu at

Doc5: ali ayşeye topu at

- Calculate the term-document matrix.
- Calculate the term-term matrix for a word window of ± 1 .
- Using the term-term matrix in b., calculate the cosine similarity between words ali and ayşe.

V. QUESTION

In Turkish morphological disambiguation, there are two major problems. The first problem is to identify the correct root forms, and the second problem is to identify the correct morphemes. Assume that we are only interested in the first problem, and assume also that you have a morphologically disambiguated corpora. Propose a root form disambiguation algorithm based on NGrams to solve that problem. Show the training algorithm.

VI. QUESTION

Given the following sentences, positive and negative categories; calculate the category for the given test sentences. Use Laplace smoothing to calculate probabilities.

Sentence	Category
hep tekrar hep tekrar	-
süper kötü bir film	-
çok iyi film	+
karakter gelişimi iyi	+
yazın en iyi film bu	+
çok süper film	?
iyi karakter çok kötü film	?

VII. QUESTION

Given two noun synsets S_1 and S_2 in the Turkish wordnet, propose an algorithm to find the shortest distance in the hypernym tree between S_1 and S_2 .